WOLF QUANT MODEL

Investment committee briefing

A disciplined global-equity decision system with portfolio constraints, stress testing and governance.

DECISION: CONTINUE PAPER AND SHADOW OPERATION

55,504	**89**	**75 / 100**	**0**
active equities	monthly decisions	governance score	hard breaches

As of 19 August 2026 | CONDITIONALLY_APPROVED | Research evidence

The decision in one page

The controls improved materially; the alpha claim did not.

Recommendation

Continue paper and shadow operation. Do not allocate live capital while validation components still fail.

What still blocks full deployment

Observed PIT evidence is incomplete, and Wolf trailed equal weight by 3.80% per year in this 60-month sample (p=0.250).

What works today

- Repeatable screening across six equity regions
- Chronological VaR governance gate: Pass
- Net-of-cost implementation controls: Warning
- Hard portfolio constraints: Pass

Decision area	Evidence	Action
Process	Checksummed	Use
Risk backtest	15 / 15	Pass
Cost controls	5 / 10	Warning
Hard constraints	10 / 10	Pass
PIT evidence	7.5 / 15	Warning
Deployable alpha	Not established	Do not claim

Approval means: Pilot capital, pre-trade review, monthly risk reporting and explicit stop conditions. It does not mean automatic execution.

What the model actually does

A monthly decision pipeline with risk controls at the end.

1 OBSERVE

Prices, filings, macro and portfolio

2 FEATURE

Point-in-time quality, value and risk signals

3 FORECAST

3, 6, 9 and 12 month distributions

4 COMPARE

Portfolio-aware, clean-sheet and LLM branches

5 CONSTRAIN

CVaR, liquidity and concentration limits

6 GOVERN

Stress, validation and committee reporting

The output is a decision package

Ranked equities, target weights, trade direction, exposure limits, risk tests, stress results and an approval status.

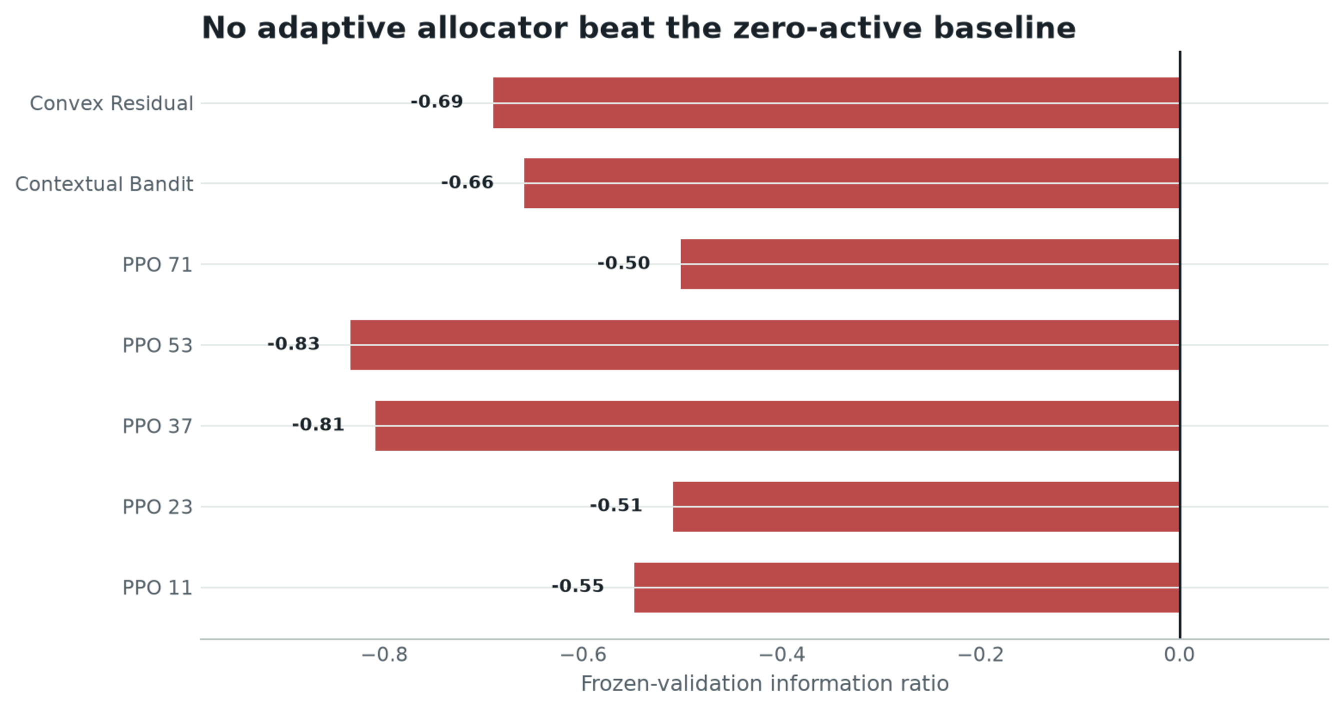
ML challengers cannot override governance

PPO, bandit, convex and supervised challengers remain research-only. Their legacy OOS results are diagnostics; the baseline remains at 100%.

Human review stays between model recommendation and execution.

DRL learned safely, but did not earn capital

Five real PPO seeds and two simpler challengers failed frozen validation.



348
regional history months

268
training months

65
validation months

12
legacy OOS months

Measured result

Best PPO validation information ratio was -0.50. 4 contiguous train-only bootstrap environments per seed produced 0 constraint violations.

Governed decision: PPO receives 0%; the classical optimiser remains 100%. That is a successful safety decision, not a failed software run.

The new supervised alpha research stack

Benchmark-relative stock ranking across four investment horizons.

1,374
research securities

16
fixed candidate specifications

4
forecast horizons: 3/6/9/12m

0%
live supervised blend

Evidence and targets

- 88 monthly decision dates
- 67 numeric plus 4 categorical features
- Returns measured versus regional and sector peers
- PIT status remains reconstructed proxy

Model challengers

- OLS with train-only Fama-MacBeth screening
- Ridge, Elastic Net and robust Huber regression
- Random Forest, Extra Trees and histogram boosting
- XGBoost return and learning-to-rank models

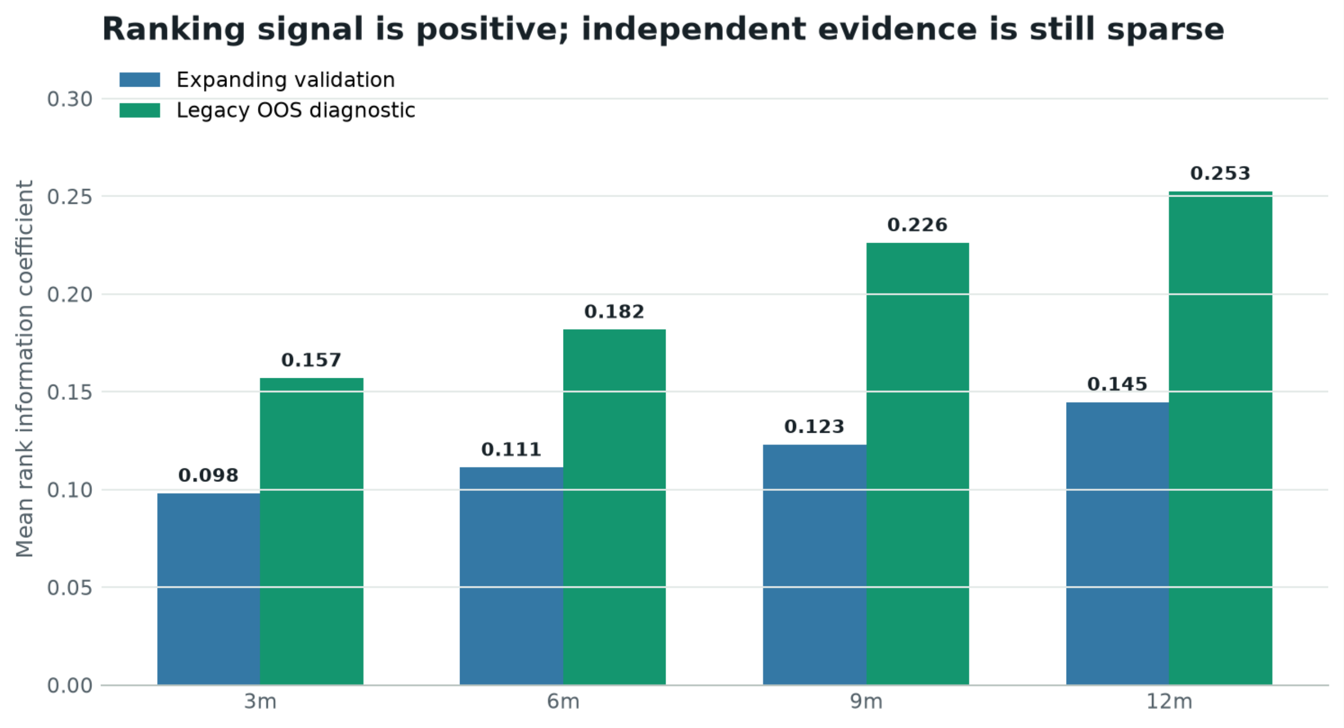
Controls before prediction

- Expanding-window purged cross-validation
- Preprocessing and screening fitted inside each fold
- One linear, tree and ranker winner per ensemble
- Explicit turnover, impact, FX and 25bp bank fee

Governance boundary: The supervised ensemble and every DRL challenger remain research-only. The governed baseline still receives 100% of portfolio weight.

Supervised signal: encouraging, not yet proven

Positive ranking results do not overcome four independent 3-month samples.



Horizon	Monthly	Independent	Rank IC	Sign p	Net cohort
3m	11	4	0.157	0.062	7.4%
6m	7	2	0.182	0.250	14.2%
9m	4	1	0.226	0.500	24.8%
12m	1	1	0.253	0.500	37.0%

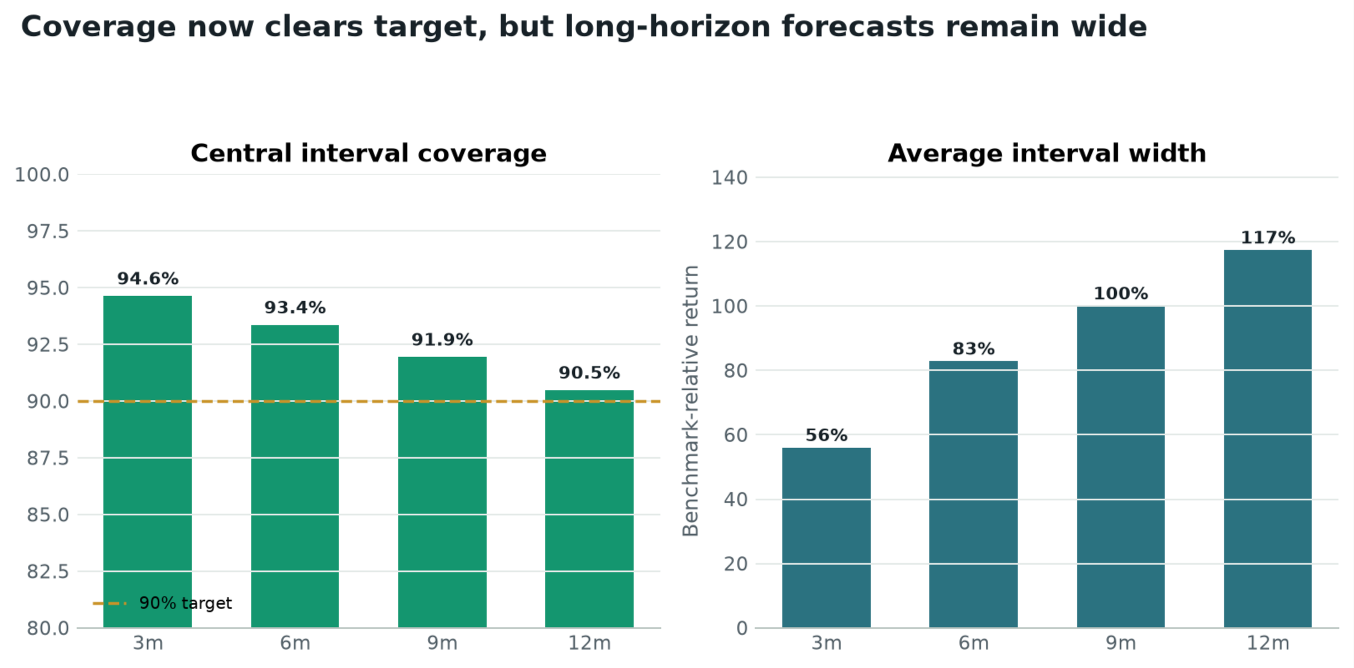
Primary 3-month diagnostic

Rank IC 0.157; mean net active cohort return 7.4%; exact sign-test p=0.0625.

What is deliberately absent: No headline CAGR, Sharpe, t-statistic or confidence interval is reported until 12 independent prospective cohorts exist.

Uncertainty and implementation are now controlled

Coverage clears target; wide long-horizon bands still signal low precision.



Horizon	Turnover	Cost drag	90% cover	Band width
3m	0.54x	0.50%	94.6%	56%
6m	0.47x	0.47%	93.4%	83%
9m	0.35x	0.42%	91.9%	100%
12m	N/A	N/A	90.5%	117%

Implementation result

Recurring turnover is 0.54x, 0.47x and 0.35x at 3/6/9 months. The 12-month record has no recurring rebalance, so it remains unestimable.

Interpretation: Calibration is fixed; precision is not. A 117% average 12-month band is a warning against confident long-horizon stock claims.

Recommendations: target portfolio versus research challengers

Only the governed CVaR target is actionable; challenger names remain shadow evidence.

20
governed target equities

20
regional challenger equities

9 / 20
names shared by both

0%
supervised live blend

Governed-only names

000001.SHE, 000538.SHE, 002352.SHE, 0823.HK, 601816.SHG, ELE.MC, EN.PA, ESSITY-B.ST, HIG.US, SEB-A.ST, SHELL.AS

Shared core

000333.SHE, 600036.SHG, AD.AS, ALV.XETRA, CA.PA, HEN3.XETRA, IMB.LSE, NOVN.SW, ORA.PA

Regional-only names

002415.SHE, 600406.SHG, CVX.US, DG.PA, DHL.XETRA, GILD.US, LOW.US, PGR.US, PUB.PA, SHEL.LSE, T.US

Highest supervised 3-month research rank in each region

Ticker	Region	Score	Cost-adj alpha	Q05	Q95
NA9.XETRA	DACH	99.6	4.2%	-33%	41%
SESG.PA	EU ex-DACH	99.2	3.7%	-31%	41%
0354.HK	Hong Kong	99.4	4.5%	-32%	43%
000009.SHE	Mainland China	85.5	2.9%	-31%	49%
GEN.LSE	UK	100.0	4.9%	-32%	39%
HPQ.US	US	100.0	3.6%	-30%	36%

Research watchlist only. These are not buy orders; uncertainty bands are wide and the deployment gate is closed.

Evidence breadth and point-in-time progress

Public identifiers, macro vintages and China/HK volume improved; historical membership remains the main gap.

55,504
active equities

1,409
walk-forward eligible

263,048
historical forecasts

48,577
current FIGI mappings

Region	Active	Master	Active share
DACH	21,603	38,792	55.7%
EU ex-DACH	2,436	5,088	47.9%
Hong Kong	3,504	4,130	84.8%
Mainland China	5,103	5,373	95.0%
UK	3,932	6,664	59.0%
US	18,926	52,523	36.0%

PIT evidence	Coverage	Status
Delisting events	59,183	Archived
Fundamentals with filing date	32,321	Mostly reconstructed
Observed acceptance times	0	Missing
Dated membership events	0	Missing
Inactive names with prices	0	Missing
Names with historical volume	8,494	Observed

Free-data checkpoint and publication boundary: yfinance volume: 5,142 China/HK names; AKShare: 12,188,157 bars across 3,374 names; FRED/ALFRED: 21 series; OpenBB: 7 benchmark checks; SEC: blocked_or_unavailable. OpenFIGI is a current snapshot, not historical membership. Only aggregate counts and checksums are published.

Equities to establish in the target portfolio

Model trade direction versus current holdings; weights are targets.

19
buy actions

1
reduce actions

5.0%
maximum name

20
equity holdings

6
equity regions

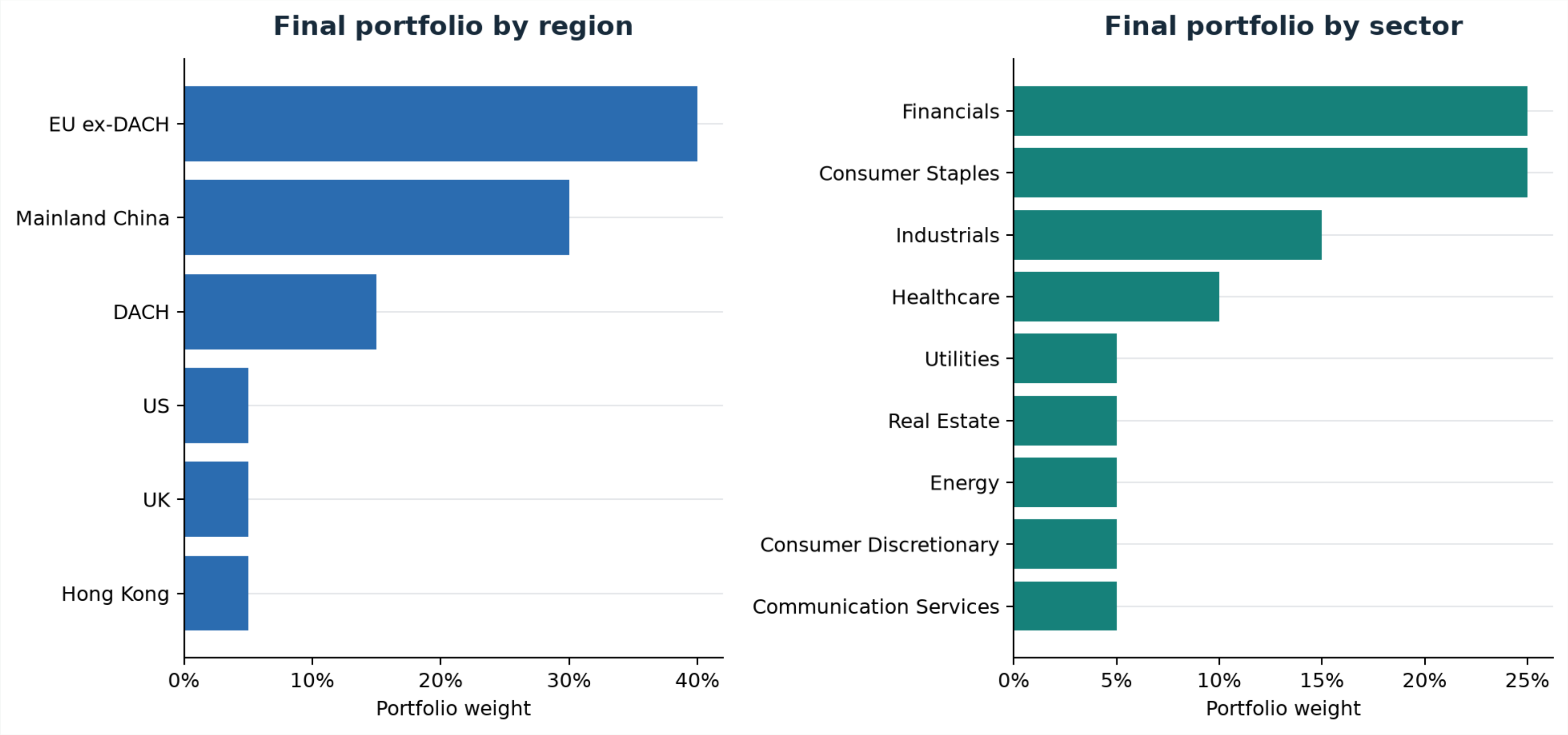
Ticker	Company	Region	Action	Target
000001.SHE	Ping An Bank Co Ltd	Mainland China	Buy	5.0%
000333.SHE	Midea Group Co Ltd	Mainland China	Buy	5.0%
000538.SHE	Yunnan Baiyao Group Co Ltd	Mainland China	Buy	5.0%
002352.SHE	S.F. Holding Co. Ltd	Mainland China	Buy	5.0%
0823.HK	Link Real Estate Investment T	Hong Kong	Buy	5.0%
600036.SHG	China Merchants Bank Co Ltd	Mainland China	Buy	5.0%
601816.SHG	Beijing-Shanghai High Speed R	Mainland China	Buy	5.0%
AD.AS	Koninklijke Ahold Delhaize NV	EU ex-DACH	Buy	5.0%
ALV.XETRA	Allianz SE VNA O.N.	DACH	Buy	5.0%
CA.PA	Carrefour SA	EU ex-DACH	Buy	5.0%

Ticker	Company	Region	Action	Target
ELE.MC	Endesa SA	EU ex-DACH	Buy	5.0%
EN.PA	Bouygues SA	EU ex-DACH	Buy	5.0%
ESSITY-B.ST	Essity AB (publ)	EU ex-DACH	Buy	5.0%
HEN3.XETRA	Henkel AG & Co. KGaA vz. (Pre	DACH	Buy	5.0%
HIG.US	Hartford Financial Services G	US	Buy	5.0%
IMB.LSE	Imperial Brands PLC	UK	Buy	5.0%
ORA.PA	Orange S.A.	EU ex-DACH	Buy	5.0%
SEB-A.ST	Skandinaviska Enskilda Banken	EU ex-DACH	Buy	5.0%
SHELL.AS	Shell PLC	EU ex-DACH	Buy	5.0%
NOVN.SW	Novartis AG	DACH	Reduce	5.0%

Execution note: recalculate orders using live NAV, FX, liquidity and prices. Manual review remains mandatory.

A deliberately diversified target

The portfolio uses equal name weights and explicit exposure caps.



Construction rules

- 20 equities at no more than 5.0% each
- 0% cash reserve
- No sector above 25%
- No country above 30%
- No region or currency above 40%
- One listing per issuer

Largest allocations

EU ex-DACH 40%; Consumer Staples 25%.
Cash is excluded from equity exposure caps.

Why the CVaR portfolio is the baseline

It balances expected return, downside loss and diversification.

Portfolio	Volatility	CVaR 5%	Yield	Hard breaches
Equal weight	14.0%	-24.9%	4.1%	0
Score weighted	12.2%	-9.4%	4.4%	0
Risk parity	11.2%	-14.9%	3.5%	0
Mean variance	11.7%	-6.9%	4.5%	0
CVaR constrained	11.8%	-6.7%	4.5%	0
regional_benchmark_relative_cost_aware	14.0%	-15.3%	4.3%	0
Dividend income	13.7%	-14.1%	5.2%	0
Regime aware	13.0%	-12.1%	5.0%	0

Selected: CVaR constrained

The baseline keeps the forecast return target while reducing the average loss in the worst 5% of modeled outcomes to 5.10%.

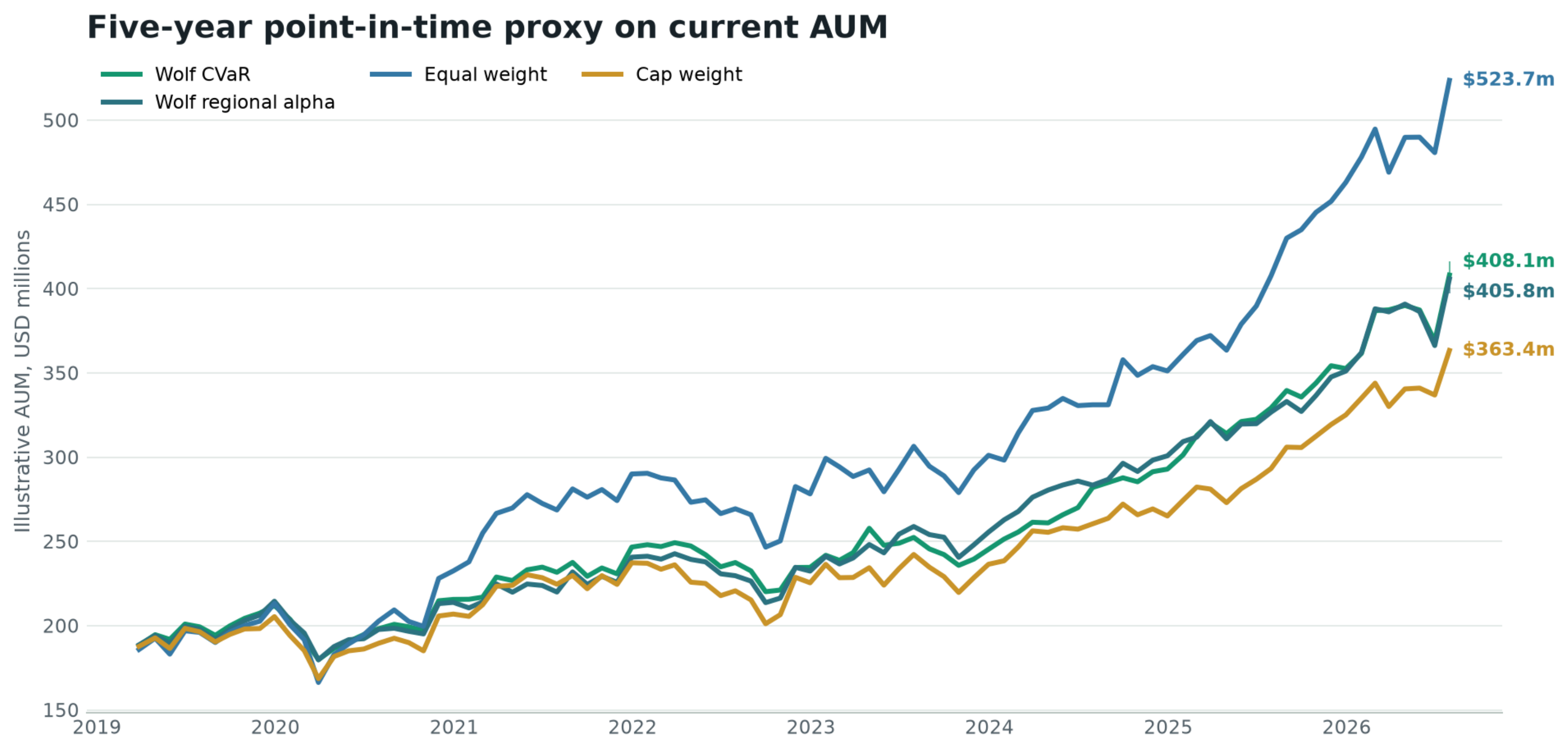
Why this matters

- Focuses on severe downside, not only average volatility
- Maintains 20 effective holdings
- Passes every hard portfolio constraint
- Produces the final 20-name equal-weight target

Interpret carefully: Expected-return inputs are model forecasts, not guaranteed market returns. Live sizing still depends on trading capacity.

Five-year point-in-time proxy

The relevant model evidence uses dated monthly decisions.



Illustrative current-AUM result

$186.1m grows to $408.1m; net PnL is $222.1m after trading costs and before the bank fee.

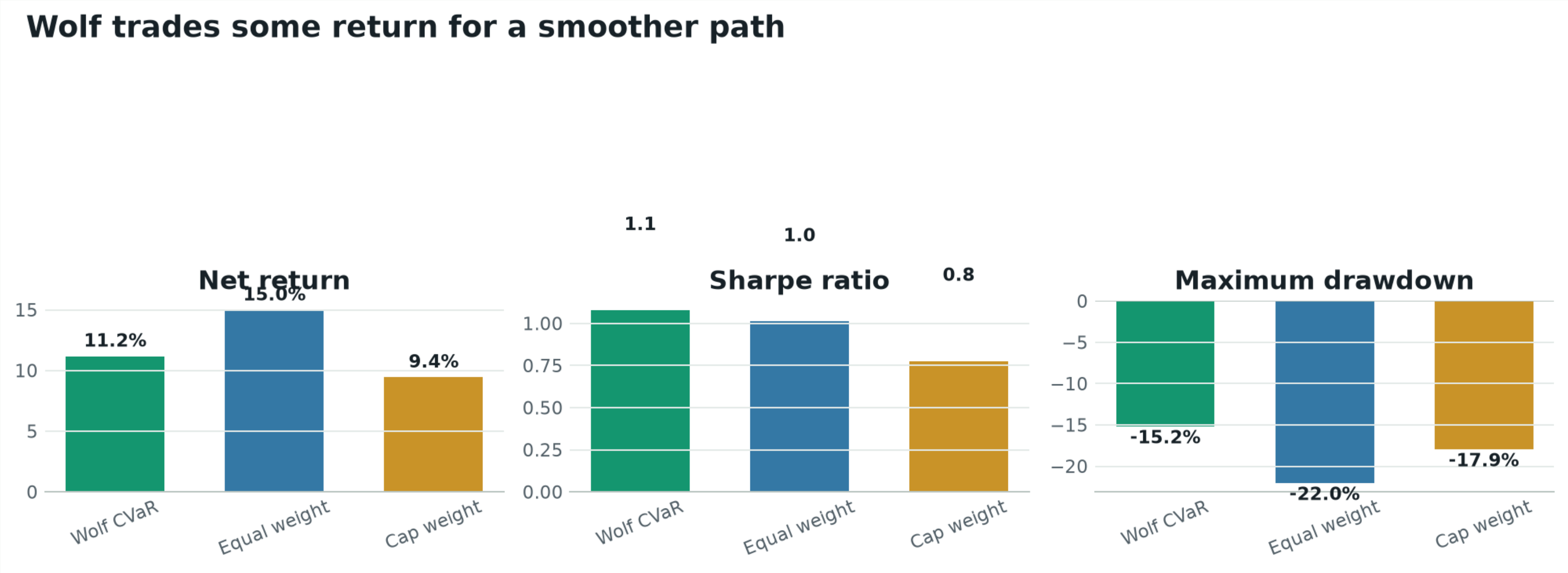
- Net annualised return: 11.2%
- Sharpe ratio: 1.08
- Maximum drawdown: -15.2%
- Positive months: 66.3%

Not a forecast

This scales the realised proxy path to current AUM. The 25 bp annual bank fee is modeled separately in the long replay.

What beat what over 60 months

Wolf delivered the best risk-adjusted result, not the best return.



The useful result

Wolf Sharpe was 1.08 and drawdown was 15.2%, better than both simple controls on these risk measures.

The honest result

Equal weight returned 14.97% versus Wolf at 11.17%. The model did not beat the simplest control on net return.

Investment interpretation

The current case is risk control and decision discipline. Incremental stock-selection alpha remains unproven.

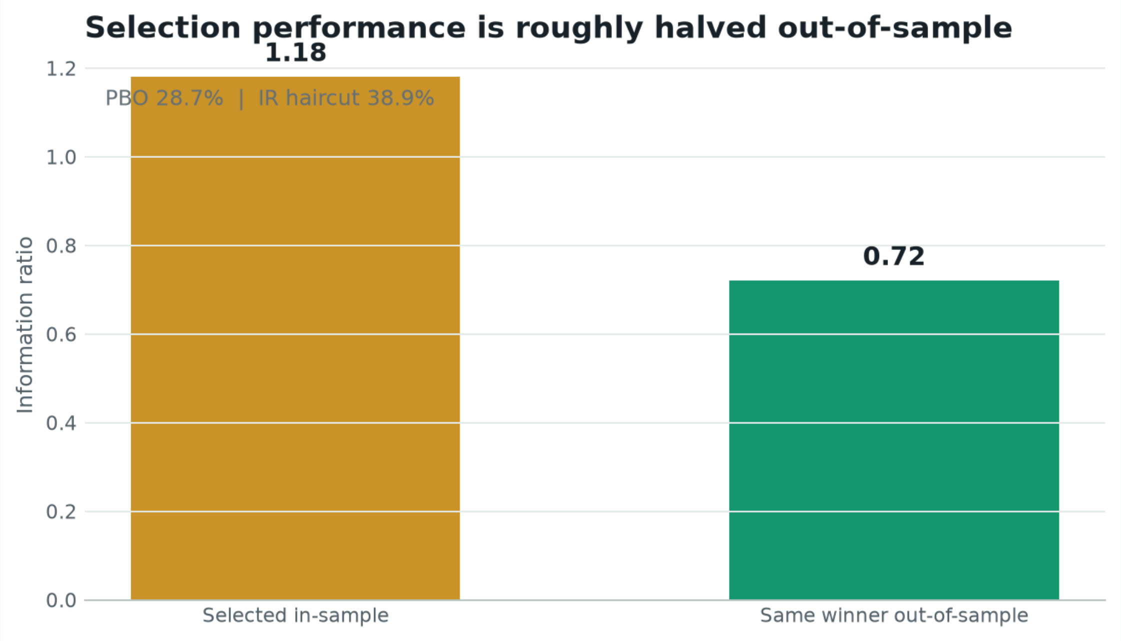
Highest return: Equal weight | Highest Sharpe: Wolf | Smallest drawdown: Wolf

Point-in-time strategy comparison, net of modeled costs

Alpha and overfitting: the hard truth

The retrospective strategies are interesting; deployable alpha is not established.

Benchmark	Active return	Reg. alpha	p-value	Verdict
Cap weight	1.38%	4.00%	0.028	NOT SIGNIFICANT
Equal weight	-3.94%	2.88%	0.212	NOT SIGNIFICANT
Wolf regional alpha	-0.01%	1.67%	0.258	NOT SIGNIFICANT



What the 50.3% fall means

When history selects the apparent winner, its information ratio typically drops from 1.32 in-sample to 0.66 out-of-sample. About half the apparent edge does not survive selection.

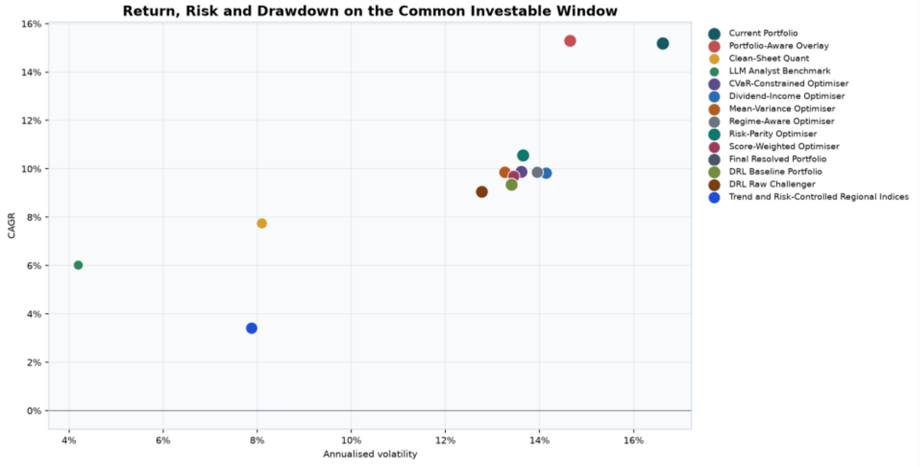
Decision

Use Wolf for disciplined ranking and risk control. Do not market it as a proven alpha engine until native live evidence clears the benchmark and cost hurdles.

The 1997 replay is a stress map

It shows how today's holdings behaved through history, not what the model knew then.

Portfolio	Start	CAGR	Sharpe	Max DD	End
Current Portfolio	$186.1m	14.1%	0.75	-42.6%	$9.0bn
Portfolio-Aware Overlay	$186.1m	13.1%	0.74	-46.4%	$7.0bn
Risk-Parity Optimiser	$100.0k	11.8%	0.79	-31.8%	$2.7m
LLM Analyst Benchmark	$100.0k	4.8%	0.66	-10.2%	$403.1k
Final Resolved Portfolio	$186.1m	9.2%	0.62	-35.8%	$2.5bn



What looks strongest

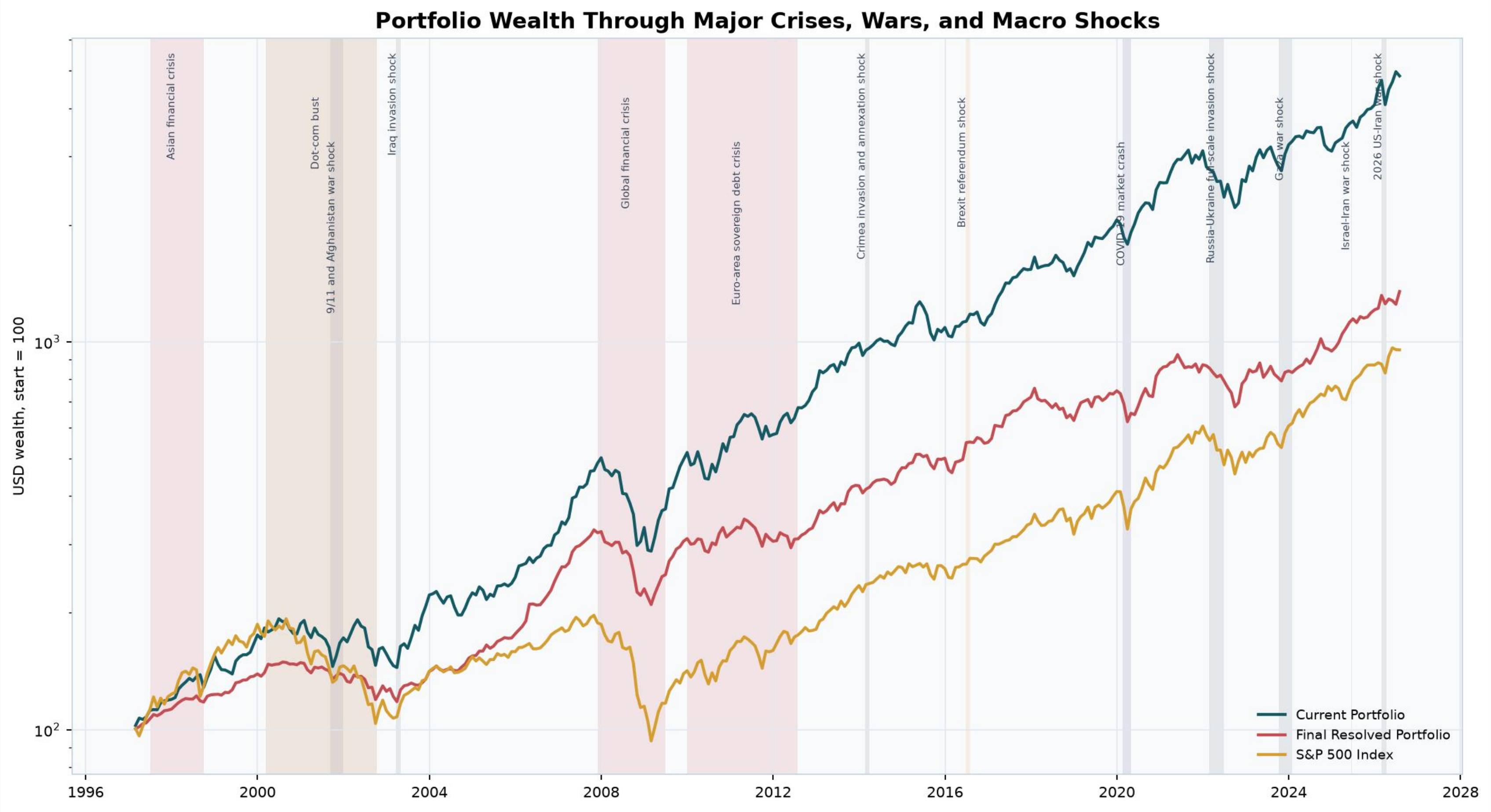
Current Portfolio PnL: $8.8bn. Final Resolved PnL: $2.3bn. Risk Parity led the independent optimisers.

Why it cannot prove selection skill

The securities were chosen with current information and then replayed backward. Survivorship and selection look-ahead make the long PnL unsuitable as a live promise.

Performance through major market shocks

Event windows are descriptive overlaps, not causal estimates.



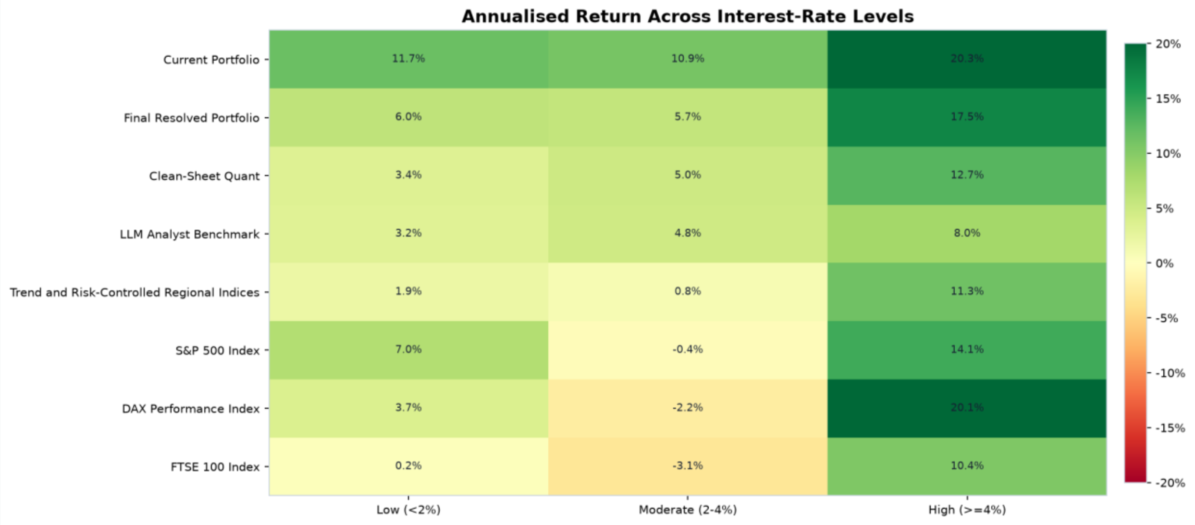
COVID-19: -12.58% return; -15.19% event drawdown

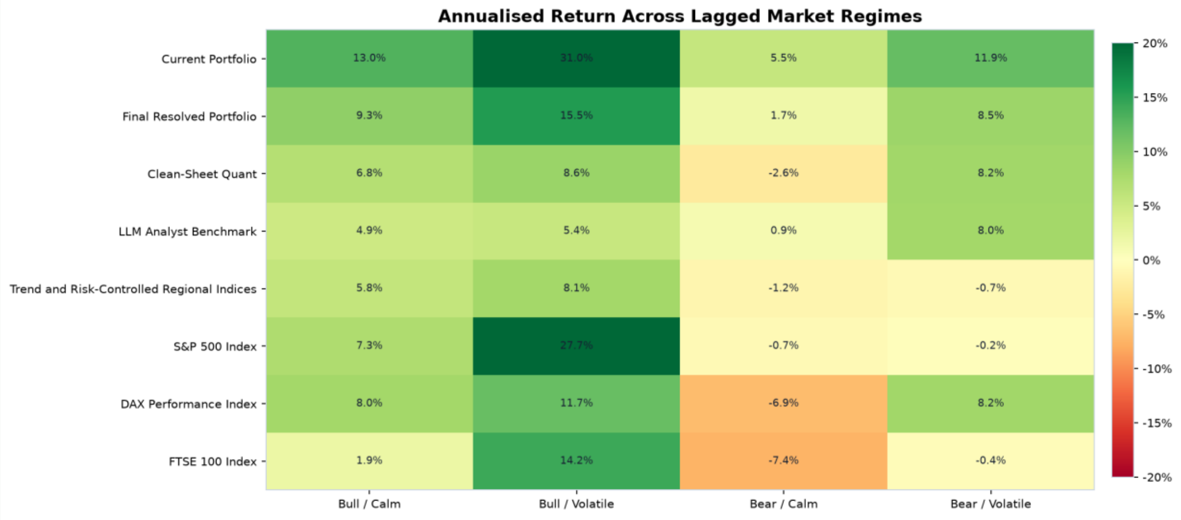
2008 crisis: -27.12% event drawdown

2026 Iran war: +3.57% return; -5.27% drawdown

Interest-rate and market-regime context

Conditional groupings show where the final portfolio was most comfortable.



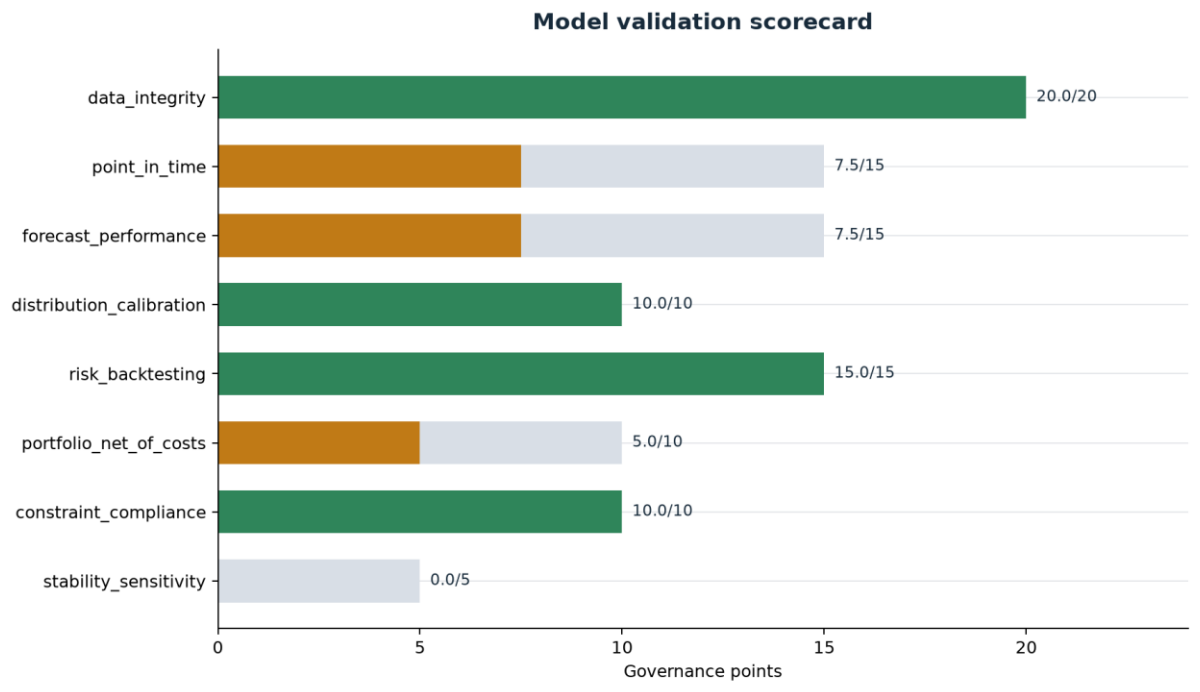


Rate reading: Final Portfolio was strongest in high-rate months (15.77%) and when rates were rising (13.03%).

Regime reading: Its strongest market group was Bull / Volatile (14.09%). Expansion months returned 9.91% annualised.

Risk holdout passes; approval stays conditional

Blocked development CV controls coverage without hiding PIT and alpha limits.



VaR coverage and independence

Sample	VaR	Exceptions	Kupiec p	Indep. p	Status
Overall	95%	98/1773	0.316	0.003	Warning
Overall	99%	28/1773	0.024	0.343	Fail
Holdout	95%	29/709	0.252	0.140	Pass
Holdout	99%	7/709	0.973	0.708	Pass

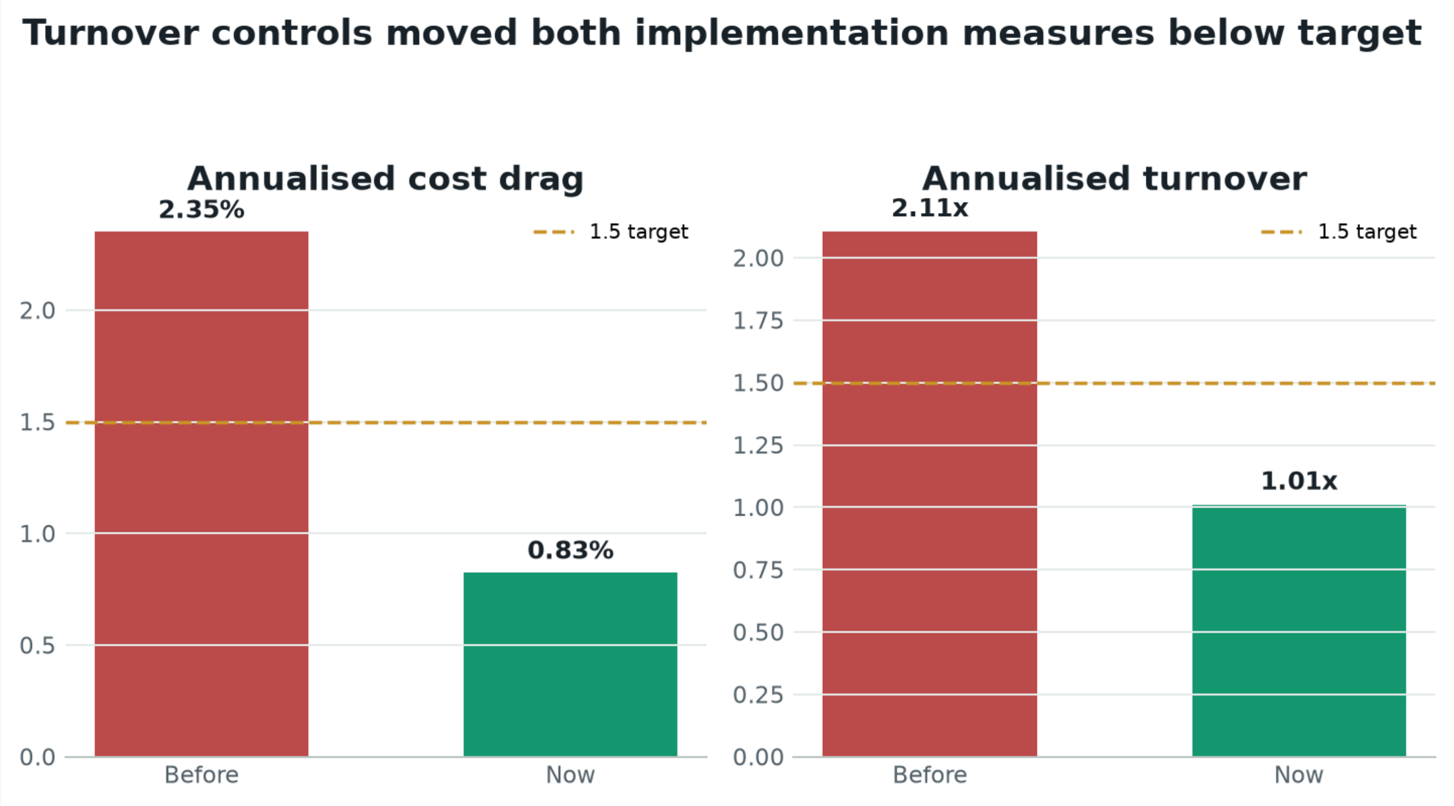
Chronological holdout gate: Pass (15 / 15)

DCC-IGARCH Student-t, filtered historical simulation, EWMA Normal and EWMA Student-t are selected using trailing data. 3 blocked development fold(s) selected a 1.000x scale and 1.20x buffer for 10 day after a breach. Overall rows remain diagnostics; the configured holdout is the gate. It is reconstructed evidence, not a pristine future shadow period.

Overall 75/100 | 4 pass | 3 warnings | 1 fail | 0 critical failures

Turnover and modeled cost drag now meet target

Retention hysteresis and minimum-turnover transitions reduced churn.



0.83%

modeled cost drag; target <= 1.50%

1.01x

annual turnover; target <= 1.50x

$465k

0.25% annual bank fee at reference AUM

Why portfolio remains a warning

Both cost gates pass, but Wolf trailed equal weight by 3.80% per year in this 60-month sample (p=0.250).

Retention and capped/minimum-turnover transitions drove the improvement; the configured no-trade band did not trigger in this sample.

A controlled path to live use

Scale only when operations, costs and evidence earn it.

0 SHADOW

4 weeks

Generate orders, do not trade. Validate data, NAV, FX and fills.

1 PILOT

6 to 12 months

Use limited capital with human approval and staged execution.

2 SCALE

Evidence gated

Increase capital only after cost, risk and benchmark hurdles hold.

Proposed pilot gates

- Zero hard constraint or data-lineage breaches
- Pre-trade cost estimate approved for every rebalance
- Keep turnover <=1.5x; stretch goal <=1.0x in the pilot
- Net performance tracked against equal and cap weight
- Monthly model-risk and investment-committee review

Immediate stop conditions

Stale critical data, any hard breach, a failed live risk test, failed reconciliation or trading costs above the approved budget.

Alpha gate: Full-scale alpha claims require native live evidence and at least 60 months under the existing governance policy.

How to read the results

Plain-language definitions for the investment committee.

Measure	Simple meaning
CAGR	Average compounded annual growth.
Sharpe	Return earned per unit of overall variability.
Sortino	Return earned per unit of harmful downside variation.
Max drawdown	Largest peak-to-trough loss.
CVaR	Average loss among the worst modeled outcomes.
Information ratio	Active return per unit of benchmark-relative risk.
p-value	How surprising the result is if true alpha were zero.
PBO	Chance that selecting the backtest winner overfits history.

Three evidence layers

1. Dated 89-month proxy: primary model evidence.
2. 1997 replay: stress and exposure history.
3. Native live record: required before full alpha approval.

Higher is not always better

A portfolio can have a higher return but a deeper drawdown, more turnover or weaker statistical evidence. The committee should judge the package, not one headline number.

Where to verify

The GitHub release contains the full PDF, plain-language interpretation, CSV evidence, plots, manifests and checksums.

THE RECOMMENDATION

Continue paper and shadow operation.

Wolf is ready to improve how decisions are made: broad screening, explicit risk, consistent portfolio construction and an audit trail. It is not yet ready to promise alpha.

USE NOW

Ranking, diversification, adaptive risk and governance

PROVE LIVE

Net benchmark edge in a genuine future shadow record

FIX NEXT

Observed PIT vintages and prospective model evidence

Investment edge today: a better-controlled process. Investment edge tomorrow: only what live evidence proves.